Information as Dependently Reconstructible Distinguishability: Carriers, Invariance Theorems, and Operational Capacity on Compact S^3 Geometry

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Abstract

Most physical accounts of information use the word before defining the object: information is said to "emerge," to be "conserved," or to be "fundamental," while the primitive it quantifies is left implicit. This paper proposes a definition-first framework. The primitive is *distinguishability*, formalized on topological carriers; the minimal carrier of an elementary distinction is S^0. We prove three structural theorems. (i) *Invariance*: the distinguishability profile $\mathsf{D}_k(X) = \operatorname{rank} \widetilde{H}_k(X)$ is preserved by every structure-preserving re-description (homotopy equivalence); elementary distinctions can be neither created nor erased invertibly. (ii) *Attributed erasure*: any change of the profile requires an explicit non-invertible operation (deletion, identification, change of category), whose logical irreversibility is the topological counterpart of Landauer's thermodynamic erasure cost. (iii) *Carrier transport*: the suspension functor shifts the profile by one degree without loss, $\mathsf{D}_{k+1}(\Sigma X) = \mathsf{D}_k(X)$; along $S^0 \to S^1 \to S^2 \to S^3$ the elementary distinction becomes the fundamental class of S^3, the minimal joint carrier of phase, direction, spin, and Hopf structure. A physical layer is then added by two postulates—carrier realization and a Planck resolution grain—yielding an exact, closed-form distinguishability capacity for a compact domain: the band-limited harmonic space on S^3_R at grain ε has dimension $N = \frac{1}{6}M(M+1)(2M+1)$ with $M = \lfloor \sqrt{1 + R^2/\varepsilon^2} \rfloor$, asymptotically $\frac{1}{3}(R/\varepsilon)^3$, in exact agreement with the Weyl law. Causal integration of this capacity is bounded below by the global accessibility time $t_{\mathrm{coh}} = \pi R/c$. The framework does not replace Shannon's theory; it supplies the alphabet that Shannon's theory presupposes. All interpretive statements—dependent designation, non-emergence, the Planck boundary as the lower limit of reconstruction rather than a beginning—are labelled as interpretation and separated from the theorems.

Keywords: theory of information; distinguishability; carriers; suspension; S^3; Planck grain; Landauer principle; compact topology; spectral capacity.

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1 Introduction

1.1 The problem: information named before it is defined

The word "information" now carries load in thermodynamics, quantum theory, cosmology, and the foundations of computation. Yet the standard quantitative theories are measure theories over *presupposed* alphabets. Shannon's theory [1] quantifies the statistics of a source once a set of distinguishable symbols is given; Kolmogorov complexity [2] quantifies description length once a machine model and a symbol set are given; quantum information theory quantifies state distinguishability once a Hilbert space is given. None of them answers the prior questions: what is a distinction, what does it take for a distinction to be physically realized, what operations preserve it, what operations destroy it and at what price, and how much distinguishability a given physical domain can carry at a given resolution.

Foundational slogans have tried to close the gap from the other side. Wheeler's "it from bit" [7] makes the bit primitive and matter derivative; Landauer's "information is physical" [4] makes the carrier primitive. Both intuitions are correct in part, and both are incomplete without a formal object that says what a "bit" is before any physics, and what physics must add before that object becomes operational.

This paper supplies such an object and develops its consequences with an explicit claim-status discipline. The primitive is *distinguishability*, carried by topological spaces; the minimal carrier of one elementary distinction is the zero-sphere $S^0 = \{-1, +1\}$. The framework then proceeds in three layers:

(I) a **mathematical layer** (Secs. 2–3), in which invariance, attributed-erasure, and carrier-transport theorems are proved;

(II) an **operational layer** (Secs. 4–5), in which two postulates—carrier realization and a resolution grain—turn the mathematics into exact capacity and accessibility statements on compact S^3 geometry;

(III) an **interpretive layer** (Sec. 8), explicitly labelled, in which the ontological reading (dependent designation, non-emergence, the Planck boundary as a limit of reconstruction) is stated without being promoted to theorem.

The companion papers of this programme supply the geometric infrastructure: the suspension-chain analysis of S^3 as minimal spin/Hopf carrier [12], the Ricci-type reconstruction flow with its no-go theorems and Planck-barrier invariant region [13], and the compact causal accessibility structure of light on S^3_R [14]. The present paper is the information-theoretic keystone of that line: it states what the other papers' carriers carry.

1.2 Claim-status convention

Label	Meaning
Theorem / Proposition	Proved under stated mathematical assumptions.
Physical anchor	Established physics cited for contact, not re-derived here.
Postulate	An operational assumption of the physical layer, explicitly adjustable.
Ansatz	A proposed quantitative form, not derived from dynamics.
Interpretation	Philosophy-of-physics reading; proves no physics.

1.3 What this paper does not claim

Boundary of the paper

This paper does not replace or amend Shannon's theory, Kolmogorov complexity, or quantum information theory; it supplies the pre-statistical object those theories presuppose. It does not derive physical law from topology; does not derive the value of ℓ_P; does not prove that the observed universe is globally S^3; does not derive gravitational entropy bounds; and does not assert any ontological statement as a theorem. Every formula in the operational layer is conditional on the stated postulates.

2 The Primitive: Distinguishability Structures

2.1 Elementary distinction

Definition 2.1 (Elementary distinction)**.** An *elementary distinction* is the two-point space $S^0 = \{-1, +1\}$ with its discrete topology. No metric, phase, direction, or causal structure is assumed at this level. The two points are not two objects located somewhere; they are the two poles of a single relational structure, *this / not-this*. Neither point carries the structure alone: removing either yields a one-point space, which is no longer a distinction.

Definition 2.2 (Distinguishability profile)**.** For a nonempty topological space X of the homotopy type of a finite CW complex, the *distinguishability profile* is

$$\mathsf{D}_k(X) \;:=\; \operatorname{rank} \widetilde{H}_k(X;\mathbb{Z}), \qquad k \geq 0, \tag{1}$$

where $\widetilde{H}_*$ denotes reduced singular homology. $\mathsf{D}_0(X)$ counts independent elementary distinctions (one less than the number of path components); D_1 counts independent cyclic (phase-type) distinctions; D_2 directional (enclosure-type) distinctions; and so on. The *total content* is $\mathsf{D}(X) = \sum_{k\geq 0} \mathsf{D}_k(X)$.

For the spheres of the suspension chain,

$$\mathsf{D}_k(S^n) = \delta_{k,n}, \qquad \mathsf{D}(S^n) = 1 : \tag{2}$$

each sphere carries exactly one unit of distinguishability, concentrated in degree n. In particular $\mathsf{D}_0(S^0) = 1$ is the formal trace of the minimal distinction, and $\mathsf{D}_3(S^3) = 1$ is the fundamental class of the compact carrier.

Remark 2.3 (Why homology). The profile could be built from other invariants. Reduced homology is chosen because it is (a) homotopy invariant, so it cannot see mere re-description; (b) computable; (c) graded, so it separates *kinds* of distinction (component / cycle / enclosure) rather than collapsing them into one number; and (d) equipped with the suspension isomorphism, which is exactly the carrier-transport operation the programme needs. Nothing in the definitions forbids refining the profile by torsion subgroups or by homotopy groups; rank homology is the minimal choice sufficient for the theorems below.

2.2 Invariance under re-description

Theorem 2.4 (Invariance of distinguishability)**.** *If $f : X \to Y$ is a homotopy equivalence, then $\mathsf{D}_k(X) = \mathsf{D}_k(Y)$ for all k, and f induces a bijection between the path components of X and Y. Consequently no structure-preserving re-description creates or erases any distinction, elementary or higher.*

Proof. A homotopy equivalence induces isomorphisms on all reduced homology groups and a bijection on π_0 [8]. Ranks and component counts are therefore equal. □

Corollary 2.5 (No invertible erasure, no invertible creation)**.** *Let $f : X \to Y$ be continuous. If f merges two path components of X (or if Y has more path components than X admits under any bijection with $\pi_0(X)$), then f is not a homotopy equivalence: elementary distinctions cannot be removed, and cannot be introduced, by any invertible operation.*

Proof. Merging components makes $\pi_0(f)$ non-injective; a component surplus makes it non-surjective. Either contradicts the bijectivity required by Theorem 2.4. □

2.3 Attributed erasure: every change of profile has an exhibited operator

Distinctions can of course be destroyed—but only by operations that leave the structure-preserving category.

Proposition 2.6 (The price of erasing the minimal distinction)**.** *The identification map $q : S^0 \to \{*\}$ satisfies $\mathsf{D}_0 : 1 \mapsto 0$ and admits no homotopy inverse. Deleting a point likewise destroys the structure (S^0 minus a point is not S^0), and there is no map at all from the nonempty S^0 to $\varnothing$. Hence the minimal distinction can be erased only by an explicit non-invertible operation—identification, deletion, or exit from the category—never by continuous deformation.*

Proof. $\{*\}$ has $\widetilde{H}_0 = 0$ while $\widetilde{H}_0(S^0) \simeq \mathbb{Z}$; homotopy equivalences preserve reduced homology, so q has no homotopy inverse. The remaining statements are immediate. □

Proposition 2.7 (Non-monotonicity: identifications are not pure erasure)**.** *Quotient operations need not decrease total content. Example: the interval $[0,1]$ is contractible, $\mathsf{D}([0,1]) = 0$, yet the identification $0 \sim 1$ produces S^1 with $\mathsf{D}_1 = 1$. Therefore "erasure" and "creation" are not global monotonicity statements; they are* per-operation bookkeeping*. Every change of the profile must be attributed to an exhibited operator, and the operator's effect must be computed, not assumed.*

Proof. $\widetilde{H}_1([0,1]) = 0$ and $\widetilde{H}_1(S^1) \simeq \mathbb{Z}$; the quotient map realizes the change. □

Remark 2.8 (Interpretation: interdependence cannot be contracted, only failed to be preserved). Proposition 2.6 is the formal content of a claim the programme states ontologically: minimal interdependent distinction is indestructible *within* operations required to preserve structure. If a description asserts that a distinction "has disappeared," the description owes an operator—a

quotient, a deletion, a change of category—and the operator's price is the loss of invertibility. Nothing in the structure-preserving language derives the disappearance; it can only be imposed as an act of forgetting.

Physical anchor 2.9 (Landauer's principle as the thermodynamic face of attributed erasure). In physics, the logically irreversible erasure of one elementary distinction (one bit) in an environment at temperature T dissipates at least $k_B T \ln 2$ of heat [3, 5]. The topological statement (erasure requires a non-invertible operator) and the thermodynamic statement (logical irreversibility has a minimum dissipation) are two faces of one discipline: *nothing erases a distinction for free.* We cite this as contact with established physics; we do not claim to derive the constant $k_B T \ln 2$ from topology.

3 Carriers and Carrier Transport

3.1 Carriers

Definition 3.1 (Carrier)**.** A *carrier* is a space on which a stated package of structures exists naturally and compatibly. A carrier is not, by that fact alone, a derivation of dynamics or matter content (the restricted usage of [12]).

3.2 The transport theorem

Theorem 3.2 (Carrier transport; degree-shift conservation)**.** *The suspension* $\Sigma X = (X \times [0,1])/(X \times \{0\},\, X \times \{1\})$ *induces natural isomorphisms*

$$\widetilde{H}_k(X) \;\simeq\; \widetilde{H}_{k+1}(\Sigma X) \qquad \text{for all } k \geq 0, \tag{3}$$

hence $\mathsf{D}_{k+1}(\Sigma X) = \mathsf{D}_k(X)$ *and the total content is preserved:* $\mathsf{D}(\Sigma X) = \mathsf{D}(X)$*. Suspension changes the carrier and the degree of every distinction; it neither creates nor destroys any.*

Proof. Mayer–Vietoris applied to the two cones covering ΣX, both contractible, with intersection homotopy equivalent to X [8]. Naturality is standard. □

Corollary 3.3 (The fundamental class of S^3 is a suspended distinction)**.** *Along* $S^0 \xrightarrow{\Sigma} S^1 \xrightarrow{\Sigma} S^2 \xrightarrow{\Sigma} S^3$,

$$\widetilde{H}_0(S^0) \simeq \widetilde{H}_1(S^1) \simeq \widetilde{H}_2(S^2) \simeq \widetilde{H}_3(S^3) \simeq \mathbb{Z}, \tag{4}$$

carrying the distinction class of S^0 *to a generator of* $H_3(S^3;\mathbb{Z})$ *[12]. The elementary distinction, the phase cycle, the directional enclosure, and the compact volume class are one invariant on four carriers.*

Remark 3.4 (Interpretation: carrier changes; generative relation persists). Theorem 3.2 is the exact form of the programme's central ontological formula: *information is not produced from non-being; it is transformed through carriers of distinguishability.* The invariant is neither simply the same (its degree, its available structures, its phenomenal unfolding all change) nor simply different (it is the image of one class under canonical isomorphisms). A new state of comprehension does not entail a new state of the invariant content.

3.3 Enrichment: what higher carriers add

Suspension preserves content but changes what the carrier can *do* with it. On S^1 the distinction acquires phase (U(1)); on S^2, direction ($\mathbb{C}P^1$, the Bloch sphere); on S^3, simultaneously: compactness, simple connectivity, the non-Abelian group SU(2), spin structure, parallelizability, and the nontrivial Hopf bundle $S^1 \hookrightarrow S^3 \to S^2$ with $c_1 = 1$. The minimality of S^3 for this joint package is a theorem of the companion paper [12] and is used here as infrastructure: S^3 is the first carrier on which a distinction can be phase-directed, spin-carried, and globally framed at once, while remaining *loop-trivial but carrier-nontrivial* ($\pi_1 = 0$, $H_3 \simeq \mathbb{Z}$).

Remark 3.5 (Qubit contact). Normalized two-level quantum states live on $S^3 \subset \mathbb{C}^2$; physical ray states on $\mathbb{C}P^1 \simeq S^2$; the projection is the Hopf map, and its nontriviality ($c_1 = 1$) is the precise sense in which phase and direction do not factorize globally [12]. Quantum information theory thus already operates on the minimal joint carrier identified by the suspension chain. This is contact, not derivation: probability rules and measurement maps are additional inputs.

4 The Operational Layer: Physical Realization of Distinguishability

The mathematical layer says what distinguishability is and how it moves between carriers. The physical layer says what it takes for a distinction to be *operational*—measurable, communicable, thermodynamically accountable. Two postulates suffice.

Postulate 1 (Carrier realization)**.** A distinction is physically realized only when represented on a carrier equipped with metric and causal structure. Without scale there is no distance; without time there is no causal relation; without causal relation there is no physically unfolded information; without distinguishability there is no observable phenomenon. Schematically:

$$\text{distinction} \;\to\; \text{carrier} \;\to\; \text{order/phase} \;\to\; \text{causal relation} \;\to\; \text{reconstructible information}. \tag{5}$$

Postulate 2 (Resolution grain)**.** Effective physical reconstruction assigns distinguishable causal-geometric states only at spatial scales $\geq \ell_P$ and temporal scales $\geq t_P = \ell_P/c$. S^0 is the formal carrier of minimal distinction; ℓ_P and t_P are its lower metric-causal realization in effective physics. Below the grain the model assigns no operationally verifiable state—a boundary of the reconstruction language, not an ontological wall.

Physical anchor 4.1 (Geometric support for Postulate 2). In the companion Ricci-type flow analysis [13], a reflecting barrier at $a = \ell_P$ defines a forward-invariant region with $|\mathrm{Rm}|^2 \leq C/\ell_P^4$, mirroring a self-adjoint Wheeler–DeWitt endpoint on $a \in [\ell_P, \infty)$ with zero probability flux through $a = \ell_P$. The same work proves that natural internal forcings cannot control the singular round mode, so the grain enters as a *boundary of the effective state space*, not as a smooth regulator. Postulate 2 adopts exactly this reading; it does not claim a microscopic quantum-gravity derivation of ℓ_P.

Remark 4.2 (Interpretation: probability one, not probability zero). Under Postulate 2 the effective physical domain has support on $a \geq \ell_P$. The sub-Planck interval is not another physical branch of probability one; it lies outside the assigned operational measure. The question of a "transition" from $a < \ell_P$ into physical reality does not arise inside the model, because $a < \ell_P$ is not a state of the model. Reality is not reconstructed from probability zero; the effective domain of reconstruction already has probability-one support on distinguishable causal geometry, bounded below by the Planck grain.

5 Capacity Theorems on Compact Carriers

The postulates convert the profile into countable operational alphabets. On a compact carrier the count is exact.

Definition 5.1 (Operational alphabet at grain ε)**.** Let (M, g) be a closed Riemannian manifold with Laplace–Beltrami operator Δ_g. The *band-limited space at grain* ε is

$$\mathcal{A}_\varepsilon(M, g) \;=\; \mathrm{span}\{\, \text{eigenmodes } Y : \; -\Delta_g Y = \lambda Y, \; \sqrt{\lambda} \leq 1/\varepsilon \,\}, \tag{6}$$

and the *distinguishability capacity* is $N_\varepsilon(M, g) = \dim \mathcal{A}_\varepsilon(M, g)$: the number of independently specifiable field degrees of freedom resolvable at scale ε.

Theorem 5.2 (Exact capacity of the round three-sphere)**.** *On S^3_R the scalar Laplacian has eigenvalues $\lambda_n = n(n+2)/R^2$ with multiplicities $(n+1)^2$, $n = 0, 1, 2, \ldots$ [12]. Hence, with*

$$M \;=\; \left\lfloor \sqrt{1 + R^2/\varepsilon^2} \right\rfloor, \tag{7}$$

the capacity at grain ε is exactly

$$N_\varepsilon(S^3_R) \;=\; \sum_{n=0}^{M-1} (n+1)^2 \;=\; \frac{M(M+1)(2M+1)}{6} \;\sim\; \frac{1}{3}\left(\frac{R}{\varepsilon}\right)^3 \quad (R/\varepsilon \to \infty). \tag{8}$$

Proof. The condition $\sqrt{\lambda_n} \le 1/\varepsilon$ reads $n(n+2) \le R^2/\varepsilon^2$, i.e. $(n+1)^2 \le 1 + R^2/\varepsilon^2$, i.e. $n+1 \le M$. Summing the multiplicities gives $\sum_{j=1}^{M} j^2 = M(M+1)(2M+1)/6$. The asymptotic follows from $M \sim R/\varepsilon$. □

Remark 5.3 (Weyl consistency check). The Weyl law for a closed 3-manifold gives $N(\lambda) \sim \mathrm{Vol}(M)\,\lambda^{3/2}/(6\pi^2)$. With $\mathrm{Vol}(S^3_R) = 2\pi^2 R^3$ and $\lambda = 1/\varepsilon^2$ this yields $N \sim \frac{1}{3}(R/\varepsilon)^3$, in exact agreement with (8). Two elementary sanity checks: $R/\varepsilon = 1$ gives $M = 1$, $N = 1$ (only the constant mode); $R/\varepsilon = 2$ gives $M = 2$, $N = 5$ (constant plus the four $n = 1$ modes).

Corollary 5.4 (Planck capacity of a quasi-local S^3 domain)**.** *Under Postulate 2 ($\varepsilon = \ell_P$), a quasi-local domain reconstructed as S^3_R carries the finite scalar-field capacity*

$$N_{\ell_P}(S^3_R) \;=\; \frac{M(M+1)(2M+1)}{6}, \qquad M = \left\lfloor \sqrt{1 + R^2/\ell_P^2} \right\rfloor \;\sim\; \frac{R}{\ell_P}, \tag{9}$$

so $N_{\ell_P} \sim \frac{1}{3}(R/\ell_P)^3$. For a binary source on this alphabet the classical capacity is N_{ℓ_P} bits; for an alphabet of q resolvable amplitudes per mode it is $N_{\ell_P} \log_2 q$ bits. Finiteness is a theorem given the postulates; the identification of ε with ℓ_P is the postulate, not the theorem.

Remark 5.5 (Volume scaling versus area scaling: an honest tension). Equation (8) is a *mode count of band-limited fields* and scales with volume, $(R/\ell_P)^3$. Gravitational entropy bounds suggest area scaling, $(R/\ell_P)^2$ [9]; the companion suspension paper adopts area counting for its entropy hierarchy for exactly that reason [12]. The two counts answer different questions—"how many field degrees of freedom fit at resolution ε" versus "how much entropy can gravitate before collapse"—and their reconciliation (which degrees of freedom are physical in a gravitating domain) is a genuine open problem of the framework, listed in Sec. 10. We flag the tension rather than hide it.

Proposition 5.6 (Causal integration bound)**.** *On the compact causal spacetime $(S^3_R \times \mathbb{R},\ g_R - c^2dt^2)$, the causal diameter is $D_c = \pi R$ and every event becomes causally accessible to every observer only after the global accessibility time*

$$t_{\mathrm{coh}} = \frac{\pi R}{c} \tag{10}$$

[14]. Hence no observer can have integrated the full capacity of Corollary 5.4 before t_{coh}: capacity is a property of the domain; integrated *information is bounded by causal accessibility.*

Definition 5.7 (Reconstructible information of a quasi-local domain)**.** The *reconstructible information* of a quasi-local domain at time T is the distinguishability realized in the operational alphabet $\mathcal{A}_\varepsilon$, restricted to the causally accessible region at T, above the grain of Postulate 2. This is a definition of the *object*; given a source distribution on the alphabet, its *quantity* is measured by the standard Shannon functional. The framework supplies the alphabet; Shannon's theory then applies unchanged.

6 Dynamics of Carriers: Reconstruction Flows

A theory of information must say not only how much a carrier holds but what happens to distinguishability when the carrier itself is deformed. The companion Ricci-type analysis [13] supplies the controlled laboratory:

(a) *Smoothing erases shape distinctions first.* On the Berger family the flow drives the anisotropy ratio $u \to 1$: shape/fibre-squashing distinctions are the first casualties of reconstruction.

(b) *Then the scale mode fails.* After rounding, the unnormalized flow collapses the remaining round mode; in reconstruction language, the variable has exhausted the distinguishable structure of the domain.

(c) *The topological class survives where the metric fails.* Throughout the smooth regime the carrier remains S^3; $H_3(S^3) \simeq \mathbb{Z}$ is untouched by metric degeneration until the description itself leaves the smooth category. Metric erasure and topological erasure are different operators with different prices (Proposition 2.6 versus curvature blow-up).

(d) *The grain closes the domain from below.* The Planck barrier makes $\{a \geq \ell_P\}$ forward-invariant with $|\mathrm{Rm}|^2 \leq C/\ell_P^4$: reconstruction never assigns states below the grain, so the informational chain (5) never evaluates on sub-grain configurations.

Remark 6.1 (Interpretation: two different limits of reconstructible information). Backward, reconstruction meets the grain: excessive curvature, $a \in [\ell_P, \infty)$. Forward, it meets not contraction but *insufficient distinguishability*: expansion, dilution, horizon formation degrade the operational alphabet accessible to any observer (Proposition 5.6 tightening against a growing domain). Past limit: the ℓ_P-boundary of smooth geometry. Future limit: loss of recoverable quasi-local structure. Neither says "reality ends"; each says this language of quasi-local informational reconstruction ceases to have full meaning there [13].

7 Relation to Existing Theories

Shannon. Shannon's theory is untouched and presupposed downstream: it measures sources on alphabets. The present framework answers the prior question of where physical alphabets come from (carrier + grain + causal access, Definition 5.7) and what operations on carriers preserve or destroy the distinctions the alphabet encodes.

Kolmogorov. Algorithmic information presupposes a symbol set and a machine; both are carriers in the present sense, and Corollary 2.5 applies to their states.

Landauer–Bennett thermodynamics of computation. Anchor 2.9: the erasure discipline proved topologically in Sec. 2 is realized thermodynamically as the $k_B T \ln 2$ bound. The two statements are logically independent and mutually reinforcing; neither is derived from the other here.

Quantum information. The qubit's phase–direction geometry is exactly the $S^3 \to S^2$ Hopf structure of the carrier chain (Sec. 3); the framework locates quantum information's state space inside the same carrier hierarchy without re-deriving quantum mechanics.

Wheeler. "It from bit" [7] makes the bit primitive. The present framework inverts the emphasis without discarding the insight: *no bit without carrier; no carrier without distinguishability*—and neither pole is a self-standing substance (Sec. 8). Information is not the ground of being here; it is the invariant of a relational structure that becomes physical only through (5).

8 Interpretive Layer

Everything in this section is labelled *Interpretation*: a reading discipline, not physics.

Remark 8.1 (Non-emergence). The theory does not assert the absolute emergence of information, space, or geometry. S^0 gives minimal distinguishability; suspension changes its carriers (Theorem 3.2); S^3 gives the special compact spin/Hopf carrier; the Planck scale gives the lower metric-causal boundary of physical reconstruction (Postulate 2). Information is not produced from non-being and is not destroyed inside admissible structural transformations; it changes its carrier of distinguishability. Cessation of a conditioned representation is not annihilation of the invariant; it is release from a conditioned carrier.

Remark 8.2 (Two protections). First, against self-being: no phenomenon in the chain—distinction, carrier, alphabet, observer—is posited as self-existing; each is given by a construction (cf. the tetralemma discipline [16]). Second, against substantializing dependence itself: having refuted self-being, one may not erect "Interdependence" as a new substance. The first part cuts off the self-being of the phenomenon; the second cuts off the self-being of dependence itself. In mathematical register: empty of own-nature, not empty of dependent functioning.

Remark 8.3 (The axis). *The Planck boundary is not the beginning of the universe; it is the universal lower boundary of quasi-local geometric reconstruction* [13]. In informational terms: not a first bit, but the grain below which the very notion of an operational bit is not assigned.

Core principle (Interpretation)

Information is dependently reconstructible distinguishability: an invariant of relational structure (Definition 2.2), preserved under structure-preserving re-description (Theorem 2.4), transported without loss across carriers (Theorem 3.2), destroyed only by exhibited non-invertible operators at a price (Proposition 2.6, Anchor 2.9), physically realized only through carrier, grain, and causal order (Postulates 1–2), and operationally finite on compact domains (Theorem 5.2). Not self-being, not nothing: dependently reconstructible distinguishability passing through a chain of carriers.

9 Falsifiability and Observational Contact

The mathematical layer is proved, not testable. The operational layer is testable at its postulates and at the geometric infrastructure it borrows.

(1) *Compact-topology searches.* If the quasi-local domain admits compact S^3-type reconstruction, the mode basis is (8)'s hyperspherical basis; matched-circle and correlation searches in the CMB constrain the admissible radii [10, 11]. A confirmed detection would fix the alphabet; continued nulls push R upward without falsifying the framework's mathematical core.

(2) *Log-periodic modulation.* The golden-ratio log-periodic ansatz $\omega = 2\pi/\ln\varphi$ associated with the φ-hierarchy of the wider programme has been directly tested by us against DESI DR2 BAO and Pantheon+ data with null results [15]; accordingly, no statement in the present paper depends on that ansatz, and we record its current status as unsupported. A theory of information that inherits a disfavored ansatz silently would violate its own erasure discipline.

(3) *Thermodynamic contact.* Single-bit Landauer dissipation has been verified experimentally [6]; this anchors the erasure discipline of Sec. 2 at the laboratory scale.

(4) *Grain postulate.* Postulate 2 is presently anchored geometrically [13], not verified experimentally; any operational sub-Planck distinguishability would falsify it and with it Corollary 5.4.

10 Conclusion and Open Problems

The paper set out to define information before using it. The result is a three-layer theory: a proved mathematical core (invariance, attributed erasure, carrier transport), an operational layer with exact capacity and accessibility formulas conditional on two explicit postulates, and an interpretive layer strictly quarantined from both. The contribution claimed is exactly this: a definition-first, theorem-based, falsifiability-aware account of what physical information is, where its alphabets come from, what preserves it, what destroys it and at what price, and how much of it a compact quasi-local domain can carry and causally integrate.

OP1. Categorical formulation. Formalize carriers and operations as a category with a distinguished class of forgetting morphisms; the profile D as a functor; erasure prices as a cost structure on non-invertible morphisms.

OP2. Quantum profile. Replace components by orthogonality: develop the distinguishability profile for state spaces (flag manifolds, $\mathbb{C}P^n$), with the Hopf class as the obstruction datum, and compare with Holevo capacity.

OP3. Capacity under reconstruction flow. Couple Theorem 5.2 to the Ricci-type flow of [13]: derive $dN_\varepsilon/d\tau$ along Berger and warped trajectories and identify the informational signature of the Planck barrier.

OP4. Volume–area reconciliation. Resolve the tension of Remark 5.5: determine which band-limited degrees of freedom remain physical in a gravitating domain, and whether the operational count degrades to area scaling dynamically.

OP5. Higher-degree erasure costs. Landauer's bound prices degree-0 erasure. Is there a thermodynamic price hierarchy for erasing degree-k distinctions (cycles, enclosures), e.g. in topological memories and topologically ordered media?

OP6. Tensor and spinor alphabets. Extend (8) to transverse-traceless tensor and Dirac spectra on S^3_R (the Dirac spectrum is $\pm(n+3/2)/R$ [12]) and give the full graded capacity of the carrier.

OP7. Integrated information dynamics. Combine Proposition 5.6 with realistic source models to bound how fast any internal observer can integrate the domain's capacity, and compare with horizon-entropy production rates.

The theory may fail; but if it fails, it fails by calculation—at a postulate, at a capacity formula, at an observational anchor—not by taste.

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